

Raghav Suriyashekar

Unity Gameplay & Systems Developer | Shader Programmer | Mobile Performance Specialist

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PROFESSIONAL PROFILE

Unity gameplay and systems developer with 3+ years of shipped experience across titles with a combined 110M+ downloads on Android and iOS, specializing in GPU-instanced rendering, mobile performance optimization, and bespoke gameplay architecture. Proven track record delivering high-performance systems under production constraints — from 10,000-unit crowd renderers sustaining 60+ FPS on legacy hardware to PS5 console ports reaching feature-complete milestone. Recognized for academic excellence as sole recipient of the Kingston University Best Postgraduate Project award and presenter at SIGGRAPH Asia 2025 for *Aural Fields*, a real-time ray-traced audio system (MSc, Distinction). Experienced working in Agile sprint environments, leading junior developers, and bridging engineering, art, and design teams to deliver on time.

CORE SKILLS & TECHNOLOGIES

Engines: Unity (primary), Unreal Engine **Languages:** C#, C++, HLSL / Shader Programming

Gameplay: Gameplay Systems, Gameplay Architecture, Physics Simulation, Inverse Kinematics, Procedural Generation

Rendering: GPU Instancing, Per-Instance Properties, Mesh-Based Shadows, Shader Programming, Shader Graph, Ray-Tracing

Performance: Mobile Optimization, Unity Profiler, Frame Debugger, Custom Metrics Pipelines

Tools & Pipeline: Editor Scripting, Tools Development, Git, Perforce, Jira, Agile / Scrum

Leadership: Team Management, Code Review, Cross-Discipline Collaboration, Sprint Planning, Stakeholder Reporting

WORK EXPERIENCE

Fountainhead Games Pvt. Ltd. | Consultant, Engineering

July 2024 – Present

- ▶ **Modular Crowd System:** Architected a modular 2D audience system with Bezier-curve-derived procedural layout, sustaining 10,000–15,000 simultaneous units at 60+ FPS on current mobile hardware and 30+ FPS on devices over 10 years old — with full designer control over position, spread, and density without engineering intervention.
- ▶ **Inverse Kinematics Framework:** Engineered a custom Inverse Kinematics framework for bat-and-ball contact resolution, delivering visually accurate animations that enhanced perceived realism and directly addressed player feedback on bat-ball connection quality.

Nextwave Multimedia Pvt. Ltd. | Gameplay & Tools Developer

May 2022 – May 2024

- ▶ **Shipped Titles:** Contributed to and built systems across WCC2 (100M+ downloads), WCC3 (10M+ downloads), and Cricket Blitz (1M+ downloads) on Android and iOS — working within bi-weekly Agile sprints, participating in stand-ups, and contributing to task prioritization and direction decisions in Jira.
- ▶ **CricStars — PS5 Console Port:** Led initial PS5 porting and development of CricStars to feature-complete milestone, designing and implementing 3 distinct control schemes tested with 30–40 internal play testers — covering gesture-mapped layouts and full integration with core gameplay systems including the bowling mechanic (spin and swing conditions) and the fielder system (vector-based target selection for natural ball-fielder interactions).
- ▶ **GPU-Instanced Shadow System:** Designed and optimized a mesh-based shadow system using GPU Instancing and per-instance properties, profiled and validated with Unity Profiler and Frame Debugger to sustain 60+ FPS on current mobile hardware and 30+ FPS on devices over 10 years old across visually complex scenes.
- ▶ **Crowd Rendering Solution:** Delivered a GPU-instanced crowd/audience rendering solution maintaining 60+ FPS on modern mobile and 30+ FPS on low-spec hardware with 10,000–15,000 simultaneous units — providing artists full parameter control over appearance and behavior without code changes.
- ▶ **Touch Input & Gesture System:** Designed and implemented a complete touch input and gesture system for an unreleased mobile football title, translating 2D screen gestures into collision-aware 3D world-space paths via a custom gesture-to-3D algorithm.
- ▶ **Team Lead:** Managed one junior developer on the football title — running daily stand-ups, conducting Git-based code reviews, delegating tasks, and acting as cross-discipline bridge between engineering, art, 3D assets, networking, DevOps, and design teams.

- ▶ **Custom Profiling Pipeline:** Built a custom performance metrics pipeline extracting key engine data for comparative analysis, producing management-ready reports that informed go/no-go decisions on game feature progression.
- ▶ **Custom QA Pipeline:** Engineered a custom QA testing suite enabling rapid automated level playthroughs with structured issue flagging across categories including level design, gameplay loop, difficulty, and glitches — cutting QA roundtrip time from 4–5 hours to under 30 minutes per 20-level set (an 85%+ reduction). Integrated a replication and replay system with time-scaling controls, allowing engineers and QA testers to isolate, reproduce, and analyze edge-case bugs with precision.
- ▶ **Level Layout Tools:** Engineered an editor-based object placement tool that reduced level design and iteration time by 75% — cutting a 2–3 hour manual process to under 30 minutes per layout — accelerating the playtesting cycle and enabling faster identification of unbalanced or unengaging level designs.

EDUCATION

Kingston University, London, United Kingdom

Distinction

MSc Game Development (Programming) | 2025

SRM Institute of Science and Technology, Tamil Nadu, India

8.31 CGPA

BTech Computer Science and Engineering | 2021

PUBLICATIONS & AWARDS

Aural Fields: Real-Time Ray-Traced Audio Solution — MSc Thesis, Kingston University (2025)

- ▶ Sole recipient of the Best Postgraduate Project award across the Kingston University Digital Media cohort (September 2025).
- ▶ Accepted and presented at SIGGRAPH Asia 2025, Hong Kong — one of only 3 papers selected in the 3D Audio category at one of the world's premier computing and interactive techniques conferences.